

Large Language Models



From tokens and transformers to GPT-4, Claude, and Gemini

Tokenisation

Pretraining

RLHF & Alignment

Prompting

Hallucination

Applications

What is a Large Language Model?

A neural network trained to predict and generate human language at massive scale

Core idea: Given a sequence of tokens, predict the next token. Repeat billions of times on billions of sentences. The model learns grammar, facts, reasoning, and style—all from this one task.

175B

**GPT-3
Parameters**

Released 2020

1.8T

**Est. GPT-4
Parameters**
Mixture of
Experts

~\$100M

**Frontier
Training Cost**
USD per run
(2024)



Probabilistic

Not a lookup table. Each output is sampled from a learned probability distribution over the vocabulary.



Emergent abilities

Capabilities not explicitly trained for—arithmetic, code, analogies—appear at scale. Not fully understood.



Context window

The model reads N tokens at once (4K–2M). Everything outside the window is invisible to it.



Any text task

Translation, summarisation, Q&A, coding, classification—all reformulated as next-token prediction.

STEP 1



Tokenisation — How Text Becomes Numbers

Neural networks cannot read text — they need integers. Tokenisers split text into subword chunks.

Example: "unbelievable AI" → 4 tokens

"un"

ID: 1234

"believ"

ID: 8821

"able"

ID: 3310

"Ġ AI"

ID: 9041

Byte-Pair Encoding (BPE)

Most common algorithm. Starts with characters, merges frequent pairs iteratively until vocabulary size (e.g. 50,257 for GPT-2) is reached.

Vocabulary size matters

GPT-4 uses ~100K tokens. More tokens = shorter sequences = cheaper attention. Rare words split into sub-tokens. Numbers often tokenise poorly.

Space-aware encoding

A leading space is part of the token: " cat" ≠ "cat". This is why context matters even at the character level — position carries meaning.

Tokens ≠ Words

1 token ≈ 0.75 words in English. 100 tokens ≈ 75 words. Chinese/Korean typically use more tokens per concept than English does.

The Transformer — Foundation of Every LLM

All modern LLMs (GPT, Claude, Gemini, LLaMA) are Transformer-based. The core architecture from 2017 remains intact.

Linear + Softmax → Next Token

× N layers (12–96)

Add & LayerNorm

Feed-Forward Network

Add & LayerNorm

Multi-Head Self-Attention

Token + Positional Embed

Input Tokens



Self-attention

Every token attends to every other token. $O(n^2)$ cost. Core innovation of the Transformer — captures long-range dependencies.



Depth = capacity

More layers → more abstract representations. GPT-3: 96 layers. Each layer refines the representation of every token.



Decoder-only for LLMs

GPT-style models use decoder-only Transformer. Masked self-attention prevents looking at future tokens during training.



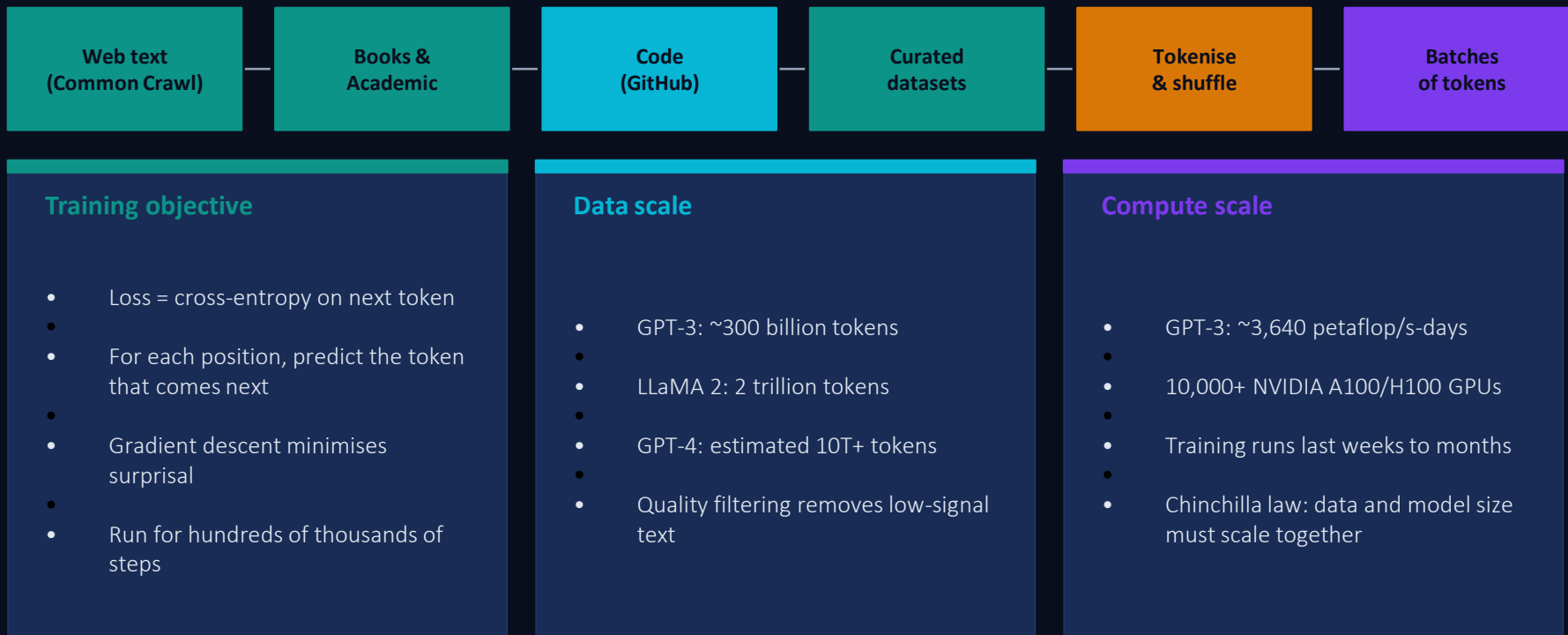
Residual connections

Skip connections add input to output of each sublayer. Prevents vanishing gradients in 96-layer deep networks.



Pretraining — Learning from the Internet

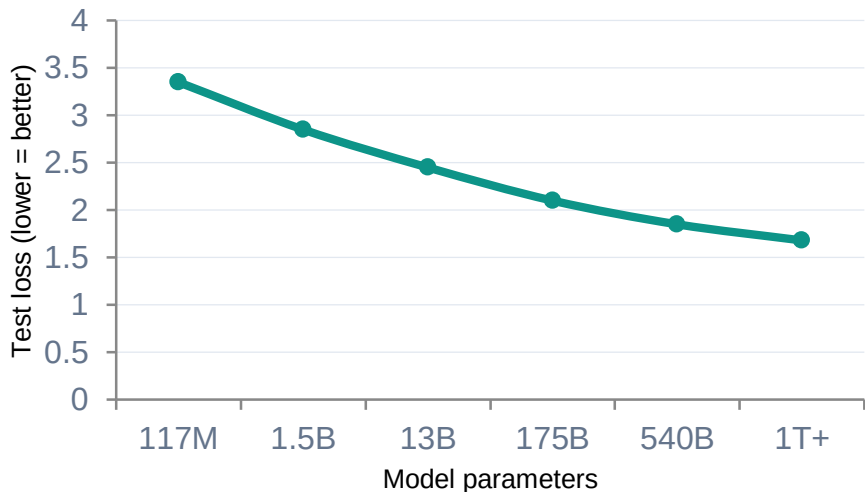
The model is trained on trillions of tokens of text using next-token prediction (causal language modelling)



Scaling Laws — Bigger Is Better (Predictably)

Loss scales as a power law with model size, dataset size, and compute — Kaplan et al. 2020, Hoffmann et al. 2022

Validation loss vs model size (schematic)



Kaplan et al. (2020)

$\text{Loss} \propto N^{(-0.076)} \cdot D^{(-0.095)} \cdot C^{(-0.050)}$. Model size, data, and compute all improve loss predictably — no sign of hitting a wall.

Chinchilla (2022)

Hoffmann et al. showed GPT-3 was under-trained. Optimal: scale model size and token count equally. Chinchilla (70B, 1.4T tokens) matched GPT-3's performance at 4× fewer parameters.

Emergent abilities

Some capabilities (arithmetic, chain-of-thought) appear suddenly at scale thresholds — not gradually. Sharp transitions are poorly understood and not predicted by loss curves.

Practical implication

Compute budget determines optimal model size AND optimal dataset size. You cannot just make the model bigger — you must also train it on more data for longer.



Fine-Tuning, RLHF & Alignment

Turning a next-token predictor into a helpful, harmless, honest assistant

1

Supervised Fine-Tuning (SFT)

Human contractors write ideal (prompt, response) pairs. Model fine-tuned on ~13K–50K examples. Teaches the assistant format and tone.

2

Reward Model Training (RM)

Human raters rank multiple model outputs. A reward model learns to predict human preference. Trained on ~33K–100K comparison pairs.

3

RL from Human Feedback (RLHF)

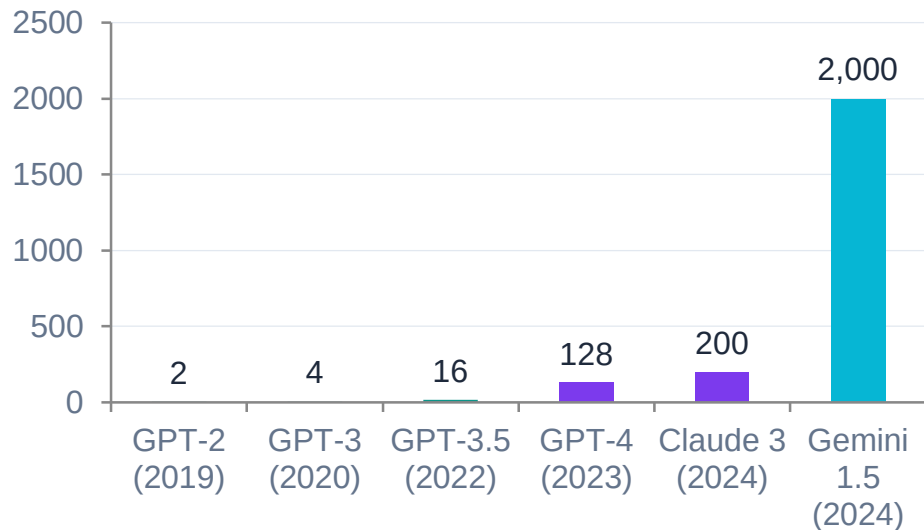
Model is updated using PPO (Proximal Policy Optimisation) to maximise reward model score. KL penalty prevents drifting too far from SFT baseline.

Constitutional AI (Anthropic): Uses a set of principles (a 'constitution') and AI self-critique to guide training — reducing reliance on human labellers and scaling alignment.

Context Window — The Model's Working Memory

The model can only see what is in its context. Outside the window = does not exist.

Context window size over time (K tokens)



Quadratic cost

Attention is $O(n^2)$ in sequence length. Doubling the context quadruples compute. Efficient attention methods (FlashAttention, sparse attention) address this.

Lost in the middle

Models perform better on information at the beginning and end of the context. Middle content is often 'forgotten'. Real limitation for long documents.

Position encoding

Rotary (RoPE) and ALiBi encodings allow extrapolation beyond training length. Naive absolute positions fail on longer-than-seen sequences.

Retrieval augmentation

RAG (Retrieval-Augmented Generation) chunks documents, embeds them, retrieves relevant chunks at inference, and injects them into the context window.



Prompting & In-Context Learning

LLMs can learn new tasks from examples in the prompt — no gradient updates required

Zero-shot

```
Prompt: "Translate to French: Hello"  
Output: "Bonjour"
```

No examples — model uses world knowledge from pretraining.

Few-shot

```
Prompt: "sad→unhappy, fast→quick, big→"  
Output: "large"
```

2–10 examples in the prompt prime the task format.

Chain-of-Thought

```
Prompt: "Think step by step: 23×17=?"  
Output: "23×17 = 23×10+23×7 = 230+161 = 391"
```

Asking the model to reason aloud improves accuracy on complex tasks by ~30%+.

System prompt

```
System: "You are a helpful coding assistant. Be  
concise."  
User: "What is a closure?"
```

Instruction-tuned models follow a system prompt that sets behaviour for the whole conversation.

LIMITATIONS

Hallucination & Critical Limitations

Understanding failure modes is as important as understanding capabilities



Hallucination

LLMs generate confident, plausible-sounding falsehoods. No internal fact-checker. The model predicts the most probable next token, not the true one. Especially dangerous in medicine and law.



No true reasoning

Models are sophisticated pattern matchers. They fail on novel multi-step problems, counting tasks, and causal reasoning that requires genuine logical inference.



Training cutoff

Knowledge frozen at training cutoff. GPT-4 cutoff: April 2023. Events after this date are unknown — the model may hallucinate or confidently answer incorrectly about recent events.



Context limits

Everything outside the context window is invisible. Long documents must be chunked. Multi-session memory requires external storage. No continuous learning during deployment.



Sycophancy

RLHF models optimise for human approval. They often agree with users even when wrong, flatter rather than correct, and change their answer if the user pushes back — even if the user is wrong.



Compute cost

GPT-3 inference: ~\$0.06 per 1K tokens (2021 prices). 1 billion queries per day would cost ~\$60M/day. Energy and hardware bottlenecks limit deployment scale.

Major LLMs — Who Made What

Key commercial and open-source large language models as of 2024

GPT-4

2023

OpenAI

Params: ~1.8T (MoE)

Context: 128K

Access: Closed

Industry standard for benchmarks; multimodal; used in ChatGPT and API

Claude 3.5

2024

Anthropic

Params: Unknown

Context: 200K

Access: Closed

Longest context; Constitutional AI; strong on coding and analysis tasks

Gemini 1.5 Pro

2024

Google DeepMind

Params: Unknown

Context: 2M

Access: Closed

Largest context window; multimodal natively; integrated into Google Workspace

LLaMA 3

2024

Meta AI

Params: 8B–70B

Context: 128K

Access: Open weights

Leading open-weight model; Apache 2.0 license; fine-tunable; runs on consumer hardware (8B)

Mixtral 8×7B

2024

Mistral AI

Params: 56B active / 8×7B

Context: 32K

Access: Open weights

Mixture of Experts (only 12.9B active per token); fast inference; rivals GPT-3.5

Qwen 2.5

2024

Alibaba

Params: 0.5B–72B

Context: 128K

Access: Open weights

Strong multilingual (Chinese/English); open weights up to 72B; competitive on coding benchmarks

Real-World Applications of LLMs

From autocomplete to autonomous agents — LLMs are reshaping every knowledge-work domain



Software Engineering

- GitHub Copilot: 46% of code written by AI (2023)
-
- Code review, test generation, debugging
-
- GPT-4 passes LeetCode hard at 46% accuracy
-
- Real-time pair programming in VS Code / JetBrains



Information Retrieval

- Bing/Google AI search integrations
-
- RAG pipelines over private document stores
-
- Perplexity.ai: LLM + web search combined
-
- Customer support chatbots replacing FAQs



Science & Research

- AlphaFold 3: protein + ligand structure prediction
-
- Drug discovery: molecule generation (Chemformer)
-
- Literature synthesis and hypothesis generation
-
- Clinical note summarisation (MIT Med)



Multimodal AI

- GPT-4V, Gemini: image understanding + text
-
- DALL-E 3, Midjourney: image generation
-
- Whisper: speech-to-text at near-human accuracy
-
- Video understanding (Gemini 1.5 Pro)



LLM Agents — Beyond Single Responses

LLMs as reasoning engines that can plan, use tools, and take multi-step actions in the world



Web search

Bing, Google, Perplexity — retrieve real-time information beyond training cutoff.



Code execution

Jupyter kernel, sandboxed Python — run generated code and observe output.



API calls

Database queries, REST APIs, calendar, email — take actions in real systems.



Memory systems

Vector stores (Pinecone, Weaviate) for long-term memory across sessions.

Ethics, Safety & Responsible Deployment

Capability is not the only concern — how we build and deploy LLMs matters enormously

Bias & fairness

LLMs amplify biases in training data. GPT-3 associated 'programmer' with male pronouns. Toxicity, stereotyping, and representational harms affect marginalised groups disproportionately.

Misinformation

Fluent, confident hallucinations at scale. LLMs are the best disinformation tools ever built — they generate convincing fake news, synthetic voices, and deepfakes at near-zero cost.

Economic disruption

McKinsey: 12M–375M workers may need to change occupations by 2030. Knowledge workers (writers, coders, paralegals) are most exposed. Historical parallels to the Industrial Revolution.

Concentration of power

Only ~5 organisations globally can train frontier models. This concentrates enormous influence over what the AI does and does not do in the hands of very few private companies.

Open Problems & Future Directions

Where the field is heading — and the unsolved problems that matter most



Reasoning & planning

Current LLMs fail on novel multi-step problems. OpenAI o1/o3, DeepSeek-R1 use chain-of-thought at inference time. True logical reasoning (not pattern matching) remains unsolved.



Multimodal & embodied

GPT-4o, Gemini: text + image + audio in one model. Next: video, 3D, robotics. Embodied LLMs that perceive and act in the physical world via sensors and actuators.



Efficient inference

H100 chips cost \$30K+ each. Quantisation (4-bit weights), mixture of experts (only 12.9B of 56B params active), speculative decoding — racing to cut inference cost by 100x.



Continual learning

Models currently freeze at training cutoff. Continual learning allows weight updates from new data without catastrophic forgetting — critical for real-time, always-up-to-date assistants.



Interpretability

Anthropic's mechanistic interpretability work: finding circuits inside models that implement concepts. Crucial for trust, safety, and debugging. Currently models are near-black-boxes.



Multiagent systems

Multiple LLMs collaborating: GPT-4 as planner, specialist models as subagents. Autogen, CrewAI, LangGraph frameworks. Scale of coordination challenges grows rapidly.

Key Takeaways

Six ideas to carry with you from this lecture

1

LLMs are next-token predictors

One objective — predict the next token — gives rise to all capabilities. Grammar, facts, code, and reasoning all emerge from this one task at scale.

5

Hallucination is structural

LLMs are not fact-checkers — they predict probable sequences. Confident-sounding errors are expected, not bugs. High-stakes domains need grounding, retrieval, and verification.

3

Pretraining + RLHF = assistant

Raw pretraining gives a 'document completer'. SFT + reward modelling + PPO turns it into a helpful, harmless assistant. Alignment is a separate step from capability.

4

Context window is working memory

The model sees only what is in its window. Long-context models (2M tokens) and RAG systems extend this, but there is always a boundary. Plan accordingly.

2

Scale drives performance

Chinchilla law: optimal compute splits equally between model size and training tokens. Bigger models on more data with more compute = better performance — predictably.

6

Agents multiply capabilities

LLM + tools + memory + planning loop = agent that can act in the world. Current frontier: multi-agent systems where specialised LLMs collaborate on complex tasks.

Practice Questions

Organised by level — from conceptual understanding to critical analysis

Conceptual

Q1: Why does next-token prediction lead to models that can write essays, translate languages, and solve maths? What is the relationship between the training objective and these emergent abilities?

Hint: Think: what does minimising cross-entropy on web text require the model to learn?

Applied

Q4: You are building a customer support chatbot using an LLM. What are three specific hallucination risks and how would you mitigate each one technically?

Hint: RAG, confidence scoring, human-in-the-loop fallback

Critical

Q5: A hospital wants to use GPT-4 for clinical decision support. List four concerns you would raise before deployment, with one concrete mitigation for each.

Hint: Hallucination, bias, explainability, data privacy, liability

Technical

Q2: Explain the three phases of RLHF (SFT → RM → PPO). Why can we not just fine-tune on human-rated examples directly without the reward model?

Hint: Consider: scalability of human rating vs reward model inference

Technical

Q3: What is the Chinchilla scaling law? If you have a fixed compute budget of 1000 GPU-days, how should you split it between model size and training tokens?

Hint: Hoffmann et al. (2022): equal scaling of N and D is optimal

Advanced

Q6: Explain why a decoder-only Transformer using causal masking is preferred over an encoder-decoder for general-purpose LLMs. What does causal masking trade off?

Hint: Compare bidirectional context vs unidirectional; inference flexibility